

CHAPTER 3

FLOW, SPACE, AND ACTIVITY RELATIONSHIPS

- In determining the requirements of a facility, three important considerations are flow, space, and activity relationships.
 - **Flow** depends on lot sizes, unit load sizes, material handling equipment and strategies, layout arrangement, and building configuration.
 - **Space** is a function of lot sizes, storage system, production equipment type and size, layout arrangement, building configuration, housekeeping and organization policies, material handling equipment, and office, cafeteria, and restroom design.
 - **Activity relationships** are defined by material or personnel flow, environmental considerations, organizational structure, continuous improvement methodology (teamwork activities), control issues, and process requirements.

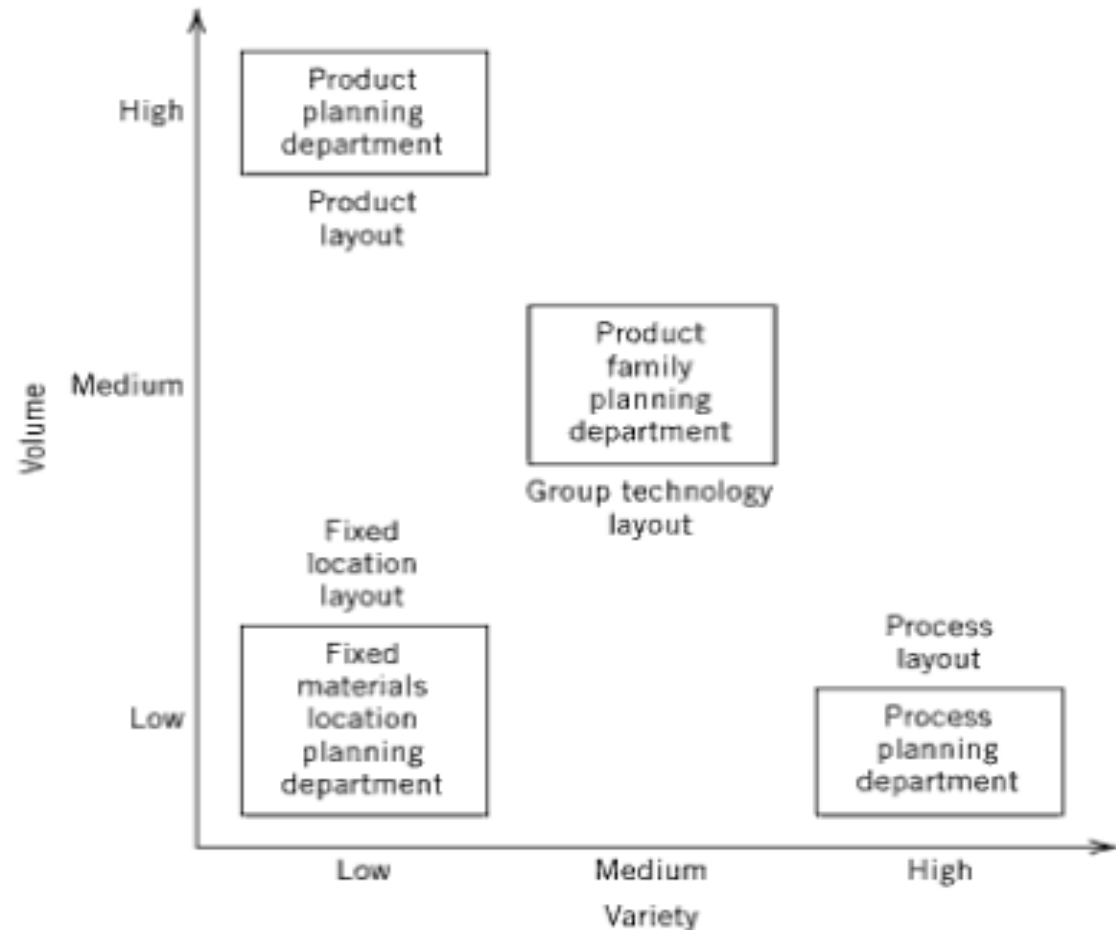
3.2 FLOW SYSTEM

- It is viewed as flow as the movement of goods, materials, energy, information and people.
- Flow may be continuous (like process industry) or discrete (like refrigeration unit).
- Flow for discrete system can be categories in
 1. Material Management System
 2. Material Flow system
 3. Physical Distribution system
- This may be combined into one overall system. Such a overall system is referred to as logistics system.

3.4 Departmental Planning

- To facilitate the consideration of flow, space, and activity relationships departmental planning is done.
- Planning departments can involve production, support, administrative, and service areas
- Production planning departments are collections of workstations to be grouped together during the facilities layout process. The formulation of organizational units should parallel the formation of planning departments.
- Depending on the product volume-variety, production planning departments can also be classified as
 1. *Product,*
 2. *Fixed materials location,*
 3. *Product family (or group technology), and*
 4. *Process planning departments*

Volume-variety layout classification



Many companies using modern manufacturing approaches are converting their facilities to combination of product and product family (group technology) planning departments. Group technology layouts are combined with *Just-In-Time (JIT)* concepts in cellular manufacturing arrangements.

If the Product Is	The Type of Planning Department Should Be	And the Method of Combining Workstations into Planning Departments Should Be
Standardized and has a large stable demand	Production line, product department	Combine all workstations required to produce the product
Physically large, awkward to move, and has a low sporadic demand	Fixed materials location, product department	Combine all workstations required to produce the product with the area required for staging the product
Capable of being grouped into families of similar parts that may be produced by a group of workstations	Product family, product department	Combine all workstations required to produce the family of products
None of the above	Process department	Combine identical workstations into initial planning departments and attempt to combine similar initial planning departments without obscuring important interrelationships within departments

Procedural guide for combining workstations in planning departments